

Toward $\mathcal{Q}_8$: a feasibility study

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Abstract

The Morin Thom polynomials are computed from a single polynomial $\mathcal{Q}_d$, the equivariant multidegree of a Borel orbit closure. $\mathcal{Q}_d$ is known for $d \leq 7$, and $d = 8$ is out of reach of Gröbner-basis methods. A formula for $\mathcal{Q}_8$ would test the conjectures of `report/ballot.pdf` at $d = 8$. We test the natural resolution candidate, the rank-one nonassociative tower, which has an explicit product formula for its point class. It agrees with $\mathcal{Q}_4$ but differs from $\mathcal{Q}_d$ in the residue-relevant sense for $d = 5, 6, 7$. Its error is a product of linear forms at $d = 5$ and then becomes irreducible: a sextic times a linear form at $d = 6$, and an irreducible polynomial of degree 13 at $d = 7$. So this route does not reach $\mathcal{Q}_8$, and no other resolution is presently in hand.

1 Setting

Let $N_d \subset \text{Hom}(\mathbb{C}^d, \text{Sym}^2 \mathbb{C}^d)$ be spanned by the coordinates q_l^{mr} with $1 \leq m \leq r$ and $m + r \leq l \leq d$. The coordinate q_l^{mr} has torus weight $z_m + z_r - z_l$. The reference point ε has $q_l^{mr} = 1$ exactly when $m + r = l$: it is the multiplication table of $t\mathbb{C}[t]/(t^{d+1})$ in the basis $t, \dots, t^d$. The upper-triangular Borel subgroup $B_d \subset GL_d$ acts on N_d . Then

$$\mathcal{Q}_d = [\overline{B_d \cdot \varepsilon}] \in \mathbb{Z}[z_1, \dots, z_d],$$

the equivariant multidegree of the orbit closure. It is homogeneous of degree $\dim N_d - \binom{d}{2} = 1, 3, 7, 13, 22$ for $d = 4, \dots, 8$. By the Bérczi–Szeneš residue formula it determines Tp_{A_d} at every relative dimension. It matters only modulo the *residue-null* polynomials, those that do not change any Chern coefficient. In this project $\mathcal{Q}_4, \dots, \mathcal{Q}_7$ are computed by Gröbner bases (SageMath/Singular) and stored as artifacts. The cost of the analogous $d = 8$ computation was estimated at $\sim 10^{11}$ times that of $d = 7$.

2 The candidate: the nonassociative tower

A resolution of the orbit closure would give $\mathcal{Q}_d$ by equivariant localisation. The natural candidate is the tower of *rank-one nonassociative* structures, built level by level. Its point class is an explicit product of weights:

$$P_d^* = \prod_{k=4}^d \prod_{\substack{2 \leq i \leq j \\ i+j \leq k}} (z_1 + z_{i-1} + z_j - z_k).$$

The number of factors is $1, 3, 7, 13, 22$ for $d = 4, \dots, 8$, matching $\deg \mathcal{Q}_d$. This class and the observation that it recovers $\mathcal{Q}_4$ are due to an unrefereed external report [1]. Everything below is recomputed here from the formula above and the stored $\mathcal{Q}_d$.

Observation 1 (verified). 1. $P_4^* = \mathcal{Q}_4$.

2. $\mathcal{Q}_d - P_d^*$ is not residue-null for $d = 5, 6, 7$. Replacing $\mathcal{Q}_d$ by P_d^* changes 21 of the 32 testable packet sums at $d = 5$, 11 of 13 at $d = 6$, and 1 of 3 at $d = 7$, on the truncations tested.
3. The differences factor over $\mathbb{Q}$ as follows:
 - at $d = 5$, $\mathcal{Q}_5 - P_5^* = (2z_1 - z_2)(z_1 + z_3 - z_4)(2z_1 + z_2 - z_5)$;
 - at $d = 6$, a linear form $(2z_1 - z_2)$ times an irreducible sextic (395 terms in all);
 - at $d = 7$, an irreducible polynomial of degree 13 (14,552 terms).

Reading. P_d^* is the class of a variety that contains the orbit closure and, from $d = 5$, has extra components. So the tower is not birational onto $\overline{B_d\epsilon}$. At $d = 5$ the defect is a single product of linear forms. That is consistent with one extra linear component, and it suggests writing $\mathcal{Q}_d$ as P_d^* minus component classes. From $d = 6$ the defect has an irreducible factor of high degree, which rules out a correction by a few linear components. At $d = 7$ nothing factors at all. The defect grows roughly like $\mathcal{Q}_d$ itself, so the tower does not reduce the problem.

3 What a route to $\mathcal{Q}_8$ would need

- **A genuinely birational smooth model of $\overline{B_d\epsilon}$** , compatible with the torus, with isolated fixed points. Localisation would then give $\mathcal{Q}_d$ as a sum over fixed points. The tower above fails this. The saturated associativity scheme has the right image but is singular.
- **Or a characterisation of $\mathcal{Q}_d$ modulo the residue-null kernel only.** Only the packet class of $\mathcal{Q}_d$ enters the Thom polynomial. Several constraints already pin that class down partially:
 - the degree;
 - the normalisation;
 - stabilisation, $[z_d^{e(d)}]\mathcal{Q}_d = \pm\mathcal{Q}_{d-1}$ with $e(d) = \lfloor d^2/4 \rfloor - d + 1$ (verified for $d \leq 7$);
 - the plane-sector theorem, which fixes every packet with $\max M \leq 1$ (given its two stated geometric inputs);
 - Rimányi’s published Tp_{A_8} at $\ell = 0$.

Whether these determine the packet class of $\mathcal{Q}_8$ is open. Counting constraints against the dimension of the space of admissible numerators is the natural next computation.

Decision. Following the gate in `report/research_plan.md`, the $\mathcal{Q}_8$ computation is not attempted. The $d = 8$ question that matters most, $\kappa_8 < 1$ in the dominance conjecture and the ballot inequality at $d = 8$, stays open.

Reproducing. The comparison uses `chernpp.optimisation.gauge.validate_gauge` on the kernel $\mathcal{Q}_d - P_d^*$, and Singular’s `factorize` on the difference.

References

- [1] Rimányi positivity: findings summary (29 July 2026), unrefereed, §8.2. Cited for the tower and the formula for P_d^* only; every claim above is recomputed.